

Observable Greenhouse Trainer

Observe, decide, actuate, present, record

A deterministic project with visible evidence and no energetic load

Lesson	024 — project	Time	240–330 minutes
Prerequisites	Lessons 001–023	Board	Arduino Mega 2560
Evidence	TP-M, D38, D13, RGB, LCD	Status	physical acceptance open
Energy class	E1: USB, passive input, LCD, and resistor-limited LEDs only		

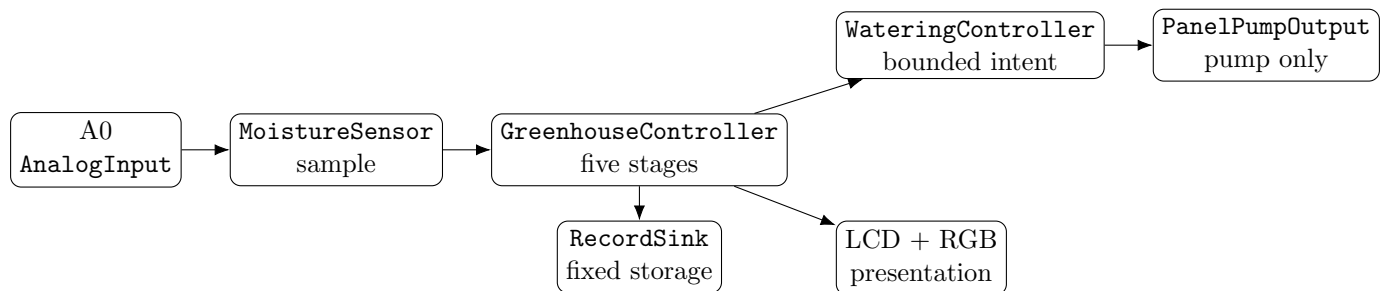
Hard boundary. “Pump” means the LED on D38. Do not connect a relay, motor, pump, heater, water, external load supply, or mains circuit. The fixed storage medium is an in-memory teaching model, not an SD card. ADK is not an unattended irrigation or safety controller.

1 Mission and dependency map

Build one replayable chain:

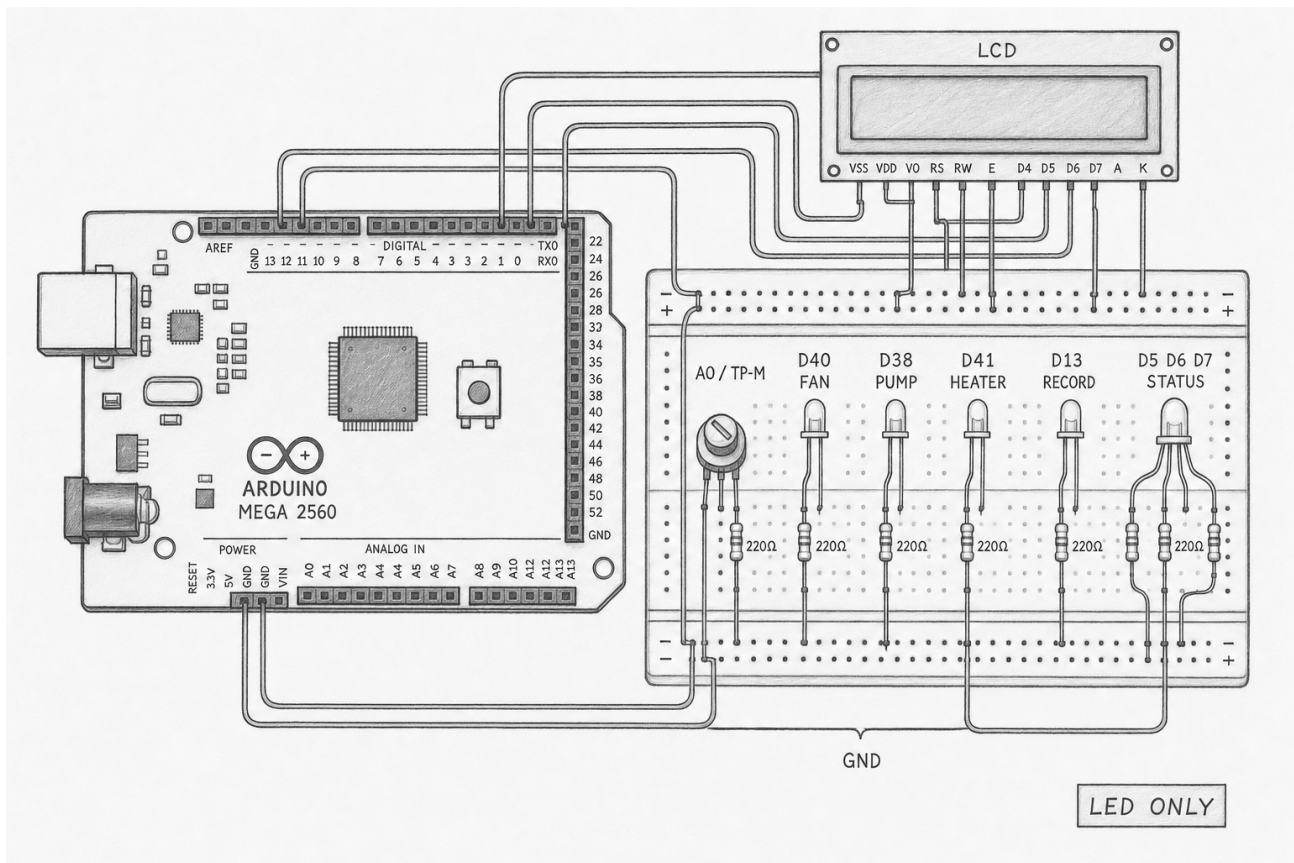
TP-M → MoistureSensor → WateringController → PanelPumpOutput → InertLoadPanel

↘ LCD view + stable record + RGB mode evidence.



The arrows are ownership or explicit borrowing relationships, not hidden global access. The coordinator owns dependency lifecycle calls; caller-owned objects outlive it.

2 Orientation and exact circuit



Alternative text: pencil orientation drawing with a Mega 2560, 10 kiloohm potentiometer on A0, 16 by 2 LCD, D40 fan, D38 pump, and D41 heater indicator LEDs, D13 record LED, and D5–D7 RGB status LED. Every external LED channel has a resistor and all paths return to common ground. The drawing contains no relay or load.

The drawing helps locate parts; this table is authoritative.

Meaning	Pin	Connection
Moisture simulator / TP-M	A0 / D54	10 k Ω potentiometer wiper
Potentiometer ends	5 V, GND	one end to each USB-powered rail
Simulated fan	D40	220 Ω , LED anode; cathode to GND
Simulated pump / TP38	D38	220 Ω , LED anode; cathode to GND
Simulated heater	D41	220 Ω , LED anode; cathode to GND
Record acknowledgement	D13	built-in LED
Mode evidence	D5, D6, D7	RGB channels, one 330 Ω resistor each
LCD control/data	D30–D35	RS, E, D4, D5, D6, D7
LCD supply	5 V, GND	exact module rating and contrast network

Unpowered inspection. Remove USB. Confirm the potentiometer endpoints, one resistor per LED channel, common ground, LCD pin order, and complete absence of a relay or external supply. Record measured resistances:

Potentiometer	_____	D40/D38/D41	_____
Red resistor	_____	Green resistor	_____
Blue resistor	_____	Board revision	_____

3 The narrative loop

```
const adk::TimePoint now (millis ());

observeGreenhouse (now);
decideGreenhouse  (now);
actuateGreenhouse (now);
presentGreenhouse (now);
recordGreenhouse  (now);
showGreenhouseHealth(now);
```

`observe()` samples at most once when due. `decide()` changes policy and intent but no output. `actuate()` is the only stage allowed to change D38. `present()` writes an already-decided LCD view. `record()` publishes the same decision. Missed periods coalesce; they do not create a burst.

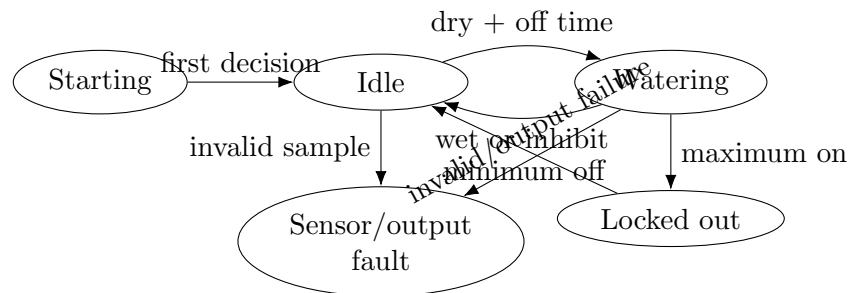
3.1 Object reading exercise

In the canonical sketch, underline each owner once and draw an arrow to every borrower. Then answer:

1. Which object claims A0? _____
2. Which object may command D38? _____
3. Which object owns the LCD pins? _____
4. Why must the record sink outlive the controller? _____
5. What runs during exception-driven unwinding in a caller? _____

4 Watering policy and state graph

The reference calibration maps raw 800 to dry (0 permille) and raw 200 to wet (1000 permille). Watering may begin below 350 only after 2000 ms off. It stops at or above 600, on inhibition, on invalid or stale sensing, or after 5000 ms on. Maximum-on creates a lockout; another dry sample cannot clear it early. `PanelPumpOutput` maps on to the panel's pump channel and off to none. The fan and heater indicators remain off throughout this project.



Condition	Decision	D38
Invalid, unavailable, or stale sample	sensor fault dominates	off
Operator inhibition	intent forced off	off
On-time at or beyond maximum	lockout	off
Valid sample at or above wet threshold	stop watering	off
Valid sample below dry threshold after off-time	begin watering	on
Between thresholds	preserve safe current state	unchanged

5 Mode precedence and independent evidence

Mode	RGB pattern	Narrow interpretation
Starting	slow blue pulse	lifecycle/schedule start
Monitoring	green heartbeat	valid sample; pump intent off
Watering	cyan heartbeat	D38 requested on
Inhibited	steady amber	software policy denied watering
SensorFault	two red pulses	sample unusable
OutputFault	steady red	command failure latched
DisplayFault	three amber pulses	LCD cannot report itself
RecordFault	four red pulses	append or sync failed
MultipleFaults	steady red	output plus another fault

Output failure dominates and latches until successful reinitialization. Sensor failure and inhibition always suppress watering. A display or record failure cannot enable D38.

5.1 Predict, observe, interpret

1. **Acquisition.** Predict the starting RGB pattern. Observe the LCD startup and RGB transition. Interpret only resource and lifecycle progress; do not infer D38 safe state.
2. **Safe state.** Predict D38 dark at startup, inhibition, invalid sensing, and shutdown. Observe the LED, then measure TP38 relative to GND. Interpret commanded level separately from resource acquisition.
3. **Record boundary.** Predict D13 changes only after append and sync succeed. Observe it at a one-second cadence. Interpret it as simulated storage acknowledgement, never SD durability.

6 Schedule and replay worksheet

Fill the prediction columns before running:

ms	raw	allowed	mode	D38	record sequence
0	500	yes	_____	_____	_____
1999	700	yes	_____	_____	_____
2000	700	yes	_____	_____	_____
4500	440	yes	_____	_____	_____
6500	700	yes	_____	_____	_____
8000	700	no	_____	_____	_____
16000	700	yes	_____	_____	_____

Replay the exact timestamps, raw values, and inhibition values twice.

Evidence	Run A digest/notes	Run B digest/notes
Snapshots	_____	_____
Pump commands	_____	_____
LCD lines	_____	_____
Record bytes	_____	_____

Conclusion: ☐ equivalent ☐ different.

If different, first divergence: _____.

7 Stable records, storage, and time

Version 1 uses bounded locale-independent ASCII:

```
adk-gh,1,17,42000,valid,472,340,on,dry,watering
```

Fields are format, version, sequence, decision milliseconds, sample state, raw reading, moisture permille or a dash, requested pump, reason, and mode. Sequence advances only after successful acceptance. One failed pending record keeps its original timestamp and bytes during retry.

`FixedStorage` distinguishes staged bytes from a durable prefix: `append()` alone is not durability; `sync()` advances the durable prefix. Shutdown and restart discard an unsynchronized suffix. This is an executable storage model in RAM, not evidence for an SD card.

The public `Rtc` boundary distinguishes valid, unset, oscillator-stopped, and transport-fault readings. The deterministic `SimulatedRtc` returns a fixed epoch plus a sequence. The record sink requires a valid reading before `append` and `sync`. No physical RTC adapter is selected, so the record schema still uses explicit `TimePoint` milliseconds.

7.1 Record decoding

Decode one observed record:

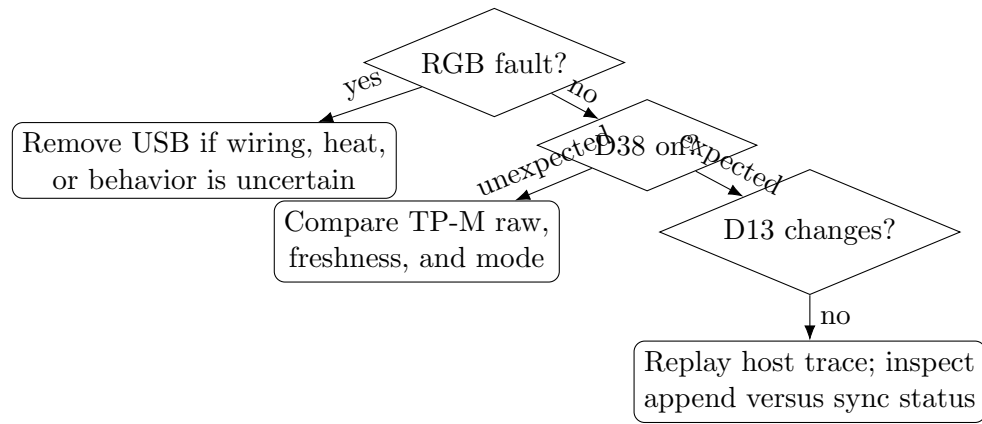
Sequence	_____	Decision ms	_____
Sample state	_____	Raw	_____
Moisture	_____	Pump	_____
Reason	_____	Mode	_____

Does the record prove physical media durability? ☐ yes ☐ no.

Explain: _____.

8 Fault matrix and diagnosis

Injected condition	Required response	Evidence to retain
Calibration invalid	initialization rejects, no output claim	complete rollback
Range or stale sample	D38 off; sensor fault visible	raw reading and state
Output command failure	best-effort off; latched output fault	full status
LCD write failure	watering state unchanged	RGB display-fault pattern
Append failure	pending record retained	original bytes and sequence
Sync failure	no durability claim; retry same record	D13 does not acknowledge
Delayed stage call	one coalesced operation	timestamp and next snapshot
Shutdown in any mode	dependencies close in reverse	TP38 safe state



Never diagnose by shorting, hot-plugging bare wires, or attaching a real load.

9 CLI experiment

```

make host-test
make host-test-sanitize
make arduino-Lesson024GreenhouseTrainer
make size-Lesson024GreenhouseTrainer
make lessons
make lessons-check
make site-check
make upload-Lesson024GreenhouseTrainer PORT=/dev/ttyACM0
make hardware-card LESSON=024 BOARD=mega2560
  
```

Record tool versions: Arduino CLI _____; AVR core _____; compiler _____.
 Firmware flash _____ bytes; static RAM _____ bytes. A compile is not a physical observation.

10 Exercises

1. Draw the exact precedence path when a record failure and stale sample coexist. Which mode appears, and why?
2. Change only the injected timestamps to cross unsigned rollover. Predict each schedule decision before replay.
3. Inject append failure, make two newer decisions, then retry. Prove the accepted bytes describe the original pending decision.
4. Design a *host-only* RTC fake trace containing Valid, NotSet, OscillatorStopped, and TransportFault. Do not invent a hardware adapter.
5. Explain why an RGB “watering” pattern plus D38 high still does not prove that a physical pump received energy.

11 Claim–evidence–reasoning

Claim	Evidence	Reasoning / limitation
Decisions replay	_____	_____
Invalid sample suppresses D38	_____	_____
Record retry is identical	_____	_____
Shutdown is safe	_____	_____

Keep claims narrow: an LED and fixed storage model cannot establish pump electrical behavior, SD durability, RTC accuracy, water delivery, plant health, or unattended safety.

12 Open Mega 2560 acceptance record

Do not complete from memory or compilation output.

Date, person, commit	_____
Board revision / USB supply	_____
Potentiometer and LCD markings	_____
Meter / logic instrument	_____
Measured 5 V rail	_____ V
TP-M at both stops	dry _____ V; wet _____ V
Measured LED resistors	D40/D38/D41 _____; RGB _____
Acquisition prediction / observation	_____
D38 safe-state prediction / observation	_____
Record acknowledgement observation	_____
Fan/pump/heater mutual exclusion	_____
Inhibition and stale-sample observation	_____
Reset and shutdown TP38 observation	_____
Deviation and interpretation	_____

Stop conditions. Remove USB on a hot part, unstable connection, voltage outside the rails, unexpected D38 state, LCD pin ambiguity, or any disagreement between table and breadboard. Software shutdown is not the power-removal method.

13 References and next step

- Canonical Lesson 024 sketch
- Controller interface and deterministic tests
- Lessons 022–024 design record and ADK safety model
- Arduino Mega 2560 Rev3 documentation
- Arduino data-logging guide

The HTML reference carries searchable API links and commands. This PDF is the printable experiment and evidence record. Next, Lesson 025 separates captured infrared timing evidence from decoded command policy.